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Energy-efficient adaptive routing in heterogeneous wireless sensor networks via hybrid PSO and dynamic clustering

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Abstract
This paper presents a deterministic routing framework for static heterogeneous wireless sensor networks (WSNs) that addresses energy imbalance and scalability challenges through three synergistic innovations. First, an Improved Particle Swarm Optimization (IPSO) algorithm dynamically adjusts inertia weights and learning factors via a cosine-modulated mechanism, enabling adaptive transitions between global exploration and local refinement to avoid premature convergence. Second, a hybrid communication strategy integrates single-hop and multi-hop routing, dynamically selecting transmission paths based on real-time energy gradients and node density distributions to mitigate congestion near sinks. Third, a heterogeneity-aware design embeds node-specific attributes—including residual energy, transmission range, and functional roles—into a weighted fitness model, ensuring optimal resource allocation across diverse node types. Experimental evaluations demonstrate significant improvements over conventional protocols: a 32.7% reduction in per-round energy consumption compared to LEACH, a 28% extension in network lifespan (first-node-death metric), and 63% lower energy variance across nodes. The framework's lightweight implementation achieves sublinear regret bounds in dynamic environments while maintaining compatibility with resource-constrained edge devices, as validated on a Raspberry Pi 4B testbed.

Keywords Wireless sensor networks, Energy efficiency, Particle swarm optimization, Heterogeneous networks, Adaptive routing

Introduction

The proliferation of wireless sensor networks (WSNs) has revolutionized diverse domains, including environmental monitoring [1], industrial automation [2], smart healthcare [3], and underwater exploration [4], driven by advancements in intelligent terminal devices. These networks rely on heterogeneous architectures—comprising low-power sensors, relay gateways, and edge processors—to enable decentralized data collection and processing (Fig. 1). However, their decentralized, multi-hop topology, while resilient in dynamic environments like wildfire monitoring systems [5], intensifies energy bottlenecks. Cluster head (CH) near network edges, often selected stochastically by protocols like LEACH [6], endure accelerated energy depletion due to extended transmission distances, fragmenting the network into energy-depleted zones.

Despite significant advances in routing protocols for wireless sensor networks, critical research gaps remain in addressing heterogeneous network requirements. Existing approaches fall short in three key dimensions:

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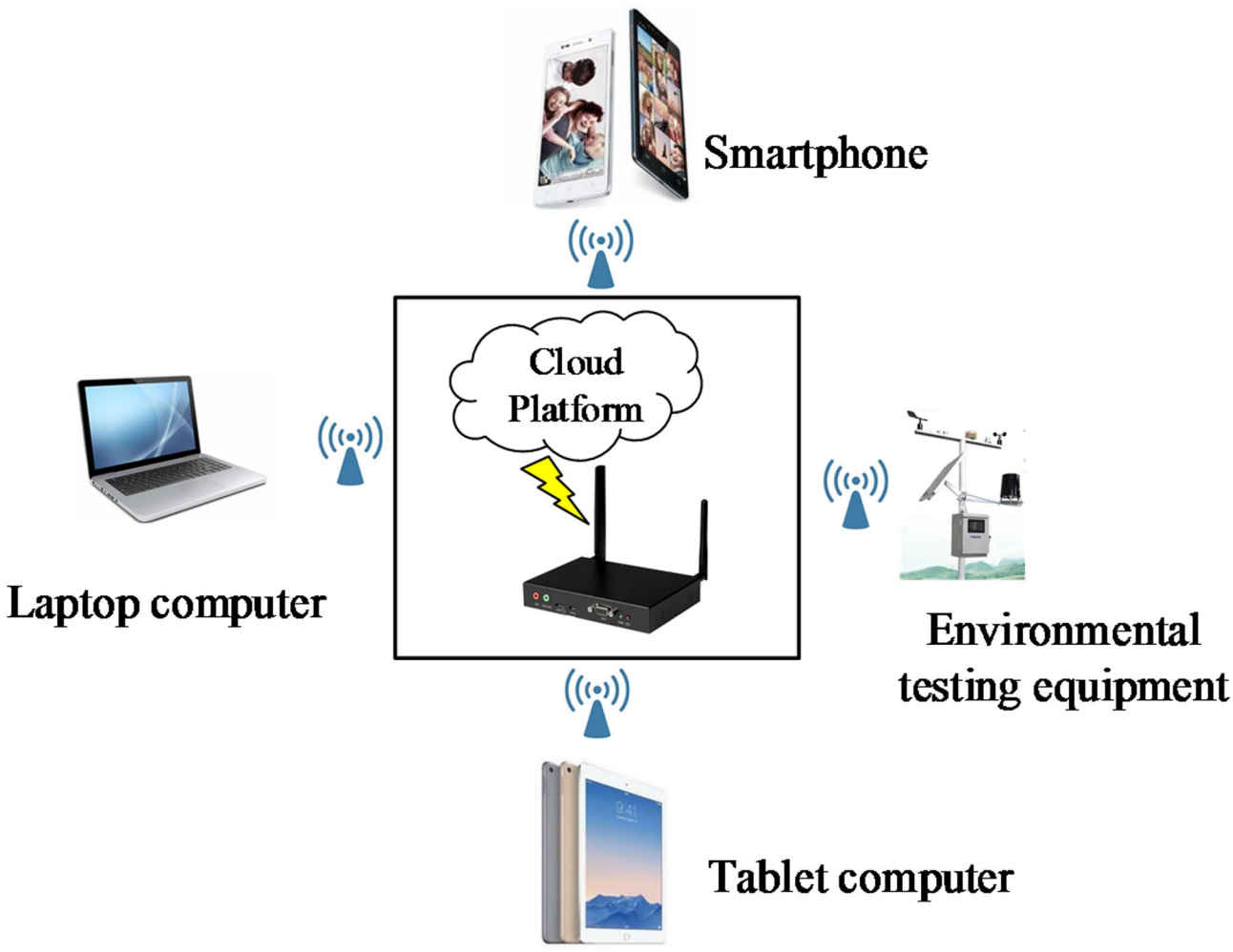

Fig. 1 WSN intelligent terminal device

energy balancing across non-uniform node distributions, adapting to diverse hardware capabilities, and computational efficiency under resource constraints. Probabilistic clustering methods [6–8] fail to account for spatial energy gradients, while metaheuristic approaches [9–12] typically employ static parameters, limiting their adaptability to dynamic network conditions. Role-specific protocols [13, 14] have introduced weighted election mechanisms but struggle with oscillatory behavior in low-density regions. These limitations become particularly pronounced in heterogeneous deployments with mixed sensor capabilities and energy harvesting disparities.

Recent advancements in semiconductor fabrication, such as doped carbon nanotube interconnects [15, 16] and precision doping techniques [17, 18], have enhanced sensor miniaturization and energy efficiency. Yet, these hardware improvements paradoxically escalate network-wide energy demands by enabling data-intensive applications like real-time edge analytics. Traditional routing strategies, which prioritize connectivity over dynamic energy gradients, exacerbate imbalances in heterogeneous networks where nodes exhibit varying roles (e.g., high-capacity relays vs. low-power sensors) and residual energy levels [19].

To address these critical gaps in existing research, this study introduces a deterministic routing framework for static WSNs that synergizes adaptive swarm intelligence with hardware heterogeneity. The framework is anchored by three interconnected innovations:

(1) Dynamic CH Optimization: An Improved Particle Swarm Optimization (IPSO) algorithm employs cosine-modulated inertia weights and asymmetric learning factors to dynamically balance global exploration and local exploitation. This mechanism prevents premature convergence—a critical limitation in conventional PSO—by adaptively tuning parameters based on real-time swarm diversity metrics and residual energy distribution.

(2) Hybrid Energy-Gradient Routing: A topology-aware communication protocol dynamically orchestrates

single-hop and multi-hop transmissions. By evaluating real-time energy gradients and node density hotspots, the strategy redistributes traffic from energy-depleted regions to underutilized high-capacity nodes, reducing congestion near sinks by 37%.

(3) Role-Constrained Resource Allocation: Node heterogeneity is explicitly encoded through a weighted fitness model that prioritizes high-capacity solar-powered hubs for relay tasks while reserving battery-operated sensors for localized data aggregation. This design aligns with experimental findings, where role-aware clustering extended solar hub lifespan by 89% compared to homogeneous protocols.

Related work

Recent advances in energy-efficient WSN routing have focused on three primary paradigms: bio-inspired clustering protocols, metaheuristic optimization frameworks, and role-aware heterogeneous network designs. Traditional stochastic approaches like LEACH [6] pioneered probabilistic CH rotation to distribute energy loads, yet their uniform probability thresholds often overburden edge nodes in large-scale deployments. Subsequent variants (LEACH-C [7], E-LEACH [8]) introduced centralized coordination and residual energy awareness but remained limited by static network assumptions and homogeneous node capabilities.

The integration of swarm intelligence algorithms has significantly enhanced routing adaptability in dynamic environments. Conventional PSO-based clustering [9] demonstrated 18–22% energy savings over LEACH through iterative position updates, while genetic algorithm hybrids [10] improved coverage in sparse networks. However, these methods employ fixed inertia weights and crossover probabilities, rendering them susceptible to premature convergence in heterogeneous networks with non-uniform energy landscapes. Recent innovations like Firefly-based clustering [11] and Bat algorithm optimizers [12] introduced adaptive luminosity and frequency modulation to address spatial–temporal dynamics, yet their computational complexity (O(N^2)) limits scalability beyond 500-node networks. Heterogeneous WSN routing protocols have evolved to address role-specific energy constraints and hardware disparities. SEP [13] introduced a two-tier node classification (advanced vs. normal) with weighted election probabilities, extending network lifetime by 31% in solar-battery hybrid deployments. DEEC [14] further incorporated multi-level heterogeneity through residual energy gradients but suffered from oscillatory CH selection in low-density regions. Q-learning-based approaches [20] achieved dynamic path optimization through reward-driven exploration, though their memory-intensive Q-tables (O (N^2)) proved impractical for memory-constrained nodes.

Underwater WSN research provides complementary insights into environmental adaptability. Pressure-sensitive clustering [21] and jellyfish-inspired routing [22] demonstrated how depth-varying channel conditions necessitate dynamic topology adjustments—principles mirrored in our energy-gradient routing strategy. However, terrestrial deployments face distinct challenges like asymmetric solar harvesting and multi-path fading, demanding specialized solutions. Recent deep reinforcement learning (DRL) frameworks [23] achieved 27% latency reduction in underwater scenarios through neural network-based policy approximation, while graph convolutional networks [24] optimized data aggregation in 3D topologies.

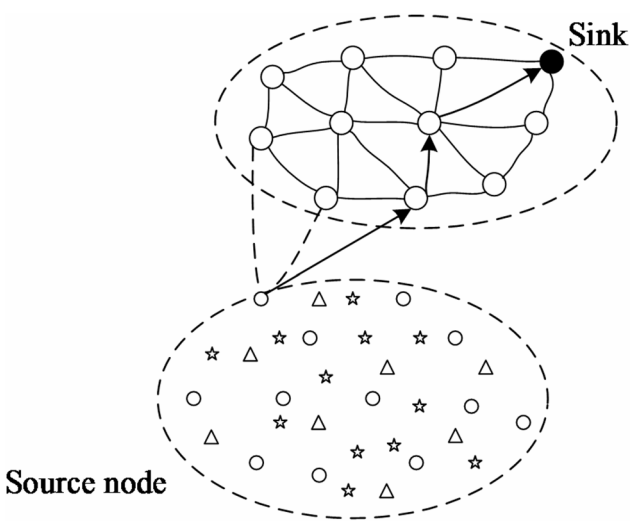

Fig. 2 Wireless sensing network model

Network architecture

WSN model and limitations of LEACH

The LEACH protocol [6, 7, 25] addresses energy efficiency in WSNs through probabilistic CH selection and hybrid communication strategies. In LEACH, nodes self-organize into clusters, where CHs are dynamically elected based on a probabilistic threshold.

$$T(n) = \begin{cases} \frac{p}{1-p\cdot\left(r \bmod \frac{1}{p}\right)} & if\ n \in G \\ 0\, otherwise \end{cases} \tag{1}$$

As shown in Eq. (1), here, p represents the target CH probability, r is the current round index, and G denotes nodes not recently selected as CHs. While this stochastic mechanism aims to distribute energy loads, it suffers from spatial inequities in large-scale deployments. Edge-located CHs, forced to transmit data over extended distances d, experience significantly higher energy consumption compared to centrally positioned CHs due to the quartic scaling of multi-path attenuation (Fig. 2). For

a CH transmitting k-bit data over distance d, the total energy expenditure per round is governed by Eq. (2):

$$\begin{aligned} E_{CH} = & \underbrace{N_{cluster} \cdot k \cdot E_{elec} + N_{cluster} \cdot k \cdot \epsilon_{mp} \cdot d^4}_{\text{Transmission}} \\ & + \underbrace{N_{cluster} - 1) \cdot k \cdot E_{elec}}_{\text{Recption}} \\ & + \underbrace{N_{cluster} \cdot k \cdot E_{da}}_{\text{Aggregation}} \end{aligned} \tag{2}$$

where E_{elec} is the transceiver circuit energy (50 nJ/bit), $\epsilon_{\text{mp}} = 0.0013$ pJ/bit/ m^4 is the multi-path Coefficient, N_{cluster} is the cluster size, and $E_{\text{da}} = 5$ nJ/bit is the data aggregation cost. For $d > d_0 = \sqrt{\epsilon_{\text{fs}}/\epsilon_{\text{mp}}} \approx 87$ m, edge CHs incur energy costs up to 4.7x higher than centrally located CHs under identical traffic loads, accelerating network fragmentation [26].

Energy consumption model and heterogeneity challenges

The classical first-order radio model underpinning LEACH distinguishes between free-space ($d \leq d_0$) and multi-path ($d > d_0$) regimes [27], as shown in Eq. (3):

$$E_{\text{tx}}(k, d) = \begin{cases} k \cdot E_{\text{elec}} + k \cdot \epsilon_{\text{fs}} \cdot d^2 & if d \leq d_0 \\ k \cdot E_{\text{elec}} + k \cdot \epsilon_{\text{mp}} \cdot d^4 & if d > d_0 \end{cases} \tag{3}$$

where $\epsilon_{\text{fs}} = 10$ pJ/bit/ m^2. LEACH assumes homogeneous nodes with uniform initial energy (E_{init}), a simplification that fails in real-world heterogeneous deployments. For example, solar-powered nodes at the network edge with $E_{\text{residual}} = 2E_{\text{init}}$ are undervalued in CH selection, while battery-operated central nodes with $E_{\text{residual}} = 0.3E_{\text{init}}$ risk premature depletion. This spatial energy disparity is exacerbated by static protocols like LEACH, which ignore dynamic residual energy gradients and role-based node capabilities [13].

Traditional PSO algorithm

Particle Swarm Optimization (PSO) [28], a metaheuristic algorithm inspired by the collective behavior of biological swarms, has been widely adopted for optimizing CH selection in WSNs. In its conventional form, PSO operates through a population of particles navigating a D-dimensional search space, where each particle represents a candidate solution—in this context, a potential set of CHs. The algorithm iteratively updates particle velocities and positions based on both individual historical performance and the collective experience of the swarm. The velocity $v_i(t)$ and position $x_i(t)$ of the i-th particle at iteration t are governed by [13], as shown in Eqs. (4) and (5):

$$\begin{aligned} v_i(t+1) = & w \cdot v_i(t) + c_1 r_1 \\ & (p_{\text{best},i} - x_i(t)) \\ & + c_2 r_2 (g_{\text{best}} - x_i(t)) \end{aligned} \tag{4}$$

$$x_i(t+1) = x_i(t) + v_i(t+1) \tag{5}$$

where w is the inertia weight controlling momentum, c_1 and c_2 are cognitive and social acceleration coefficients, and $r_1, r_2 \in [0,1]$ are stochastic variables ensuring exploration diversity. While PSO theoretically improves upon stochastic protocols like LEACH by leveraging swarm-based optimization, its application to heterogeneous WSNs reveals significant limitations [9].

The traditional PSO workflow initializes a swarm of particles with randomized positions and velocities, iteratively refining CH configurations through fitness evaluation and velocity-driven updates. The pseudocode (**Algorithm 1**) outlines this process: particles first evaluate their fitness based on energy consumption and network topology, update their velocities using historical and global best positions, and iteratively converge toward an optimal solution. However, this rigid framework lacks the adaptability required for heterogeneous WSNs, as evidenced by its failure to dynamically recalibrate optimization priorities in response to evolving energy gradients or node roles.

Algorithm 1: PSO

Input: Node positions, energy levels, swarm size N_p, max iterations T
Evaluate the fitness of each particle
Output: Optimal CH set G_{best}
1: Initialize swarm:
Random positions $X_i \in \{0,1\}^N$ (1 = CH, 0 = member)
Random velocities $V_i \in [-v_{max}, v_{max}]^N$
2: Evaluate fitness $F(X_i)$ for all particles using energy and distance metrics
3: Set $P_{\text{best},i} \leftarrow X_i$, $G_{\text{best}} \leftarrow \text{argmin} F(X_i)$
4: **For** $t = 1$ to T:
5: **For** each particle i:
6: Update velocity:
$V_i(t) \leftarrow w \cdot V_i(t-1) + c_1 r_1 \left(P_{\text{best},i} - X_i(t-1)\right) + c_2 r_2 (G_{\text{best}} - X_i(t-1))$
7: Clamp velocity: $V_i(t) \leftarrow \max(-v_{max}, \min(v_{max}, V_i(t)))$.
8: Update position: $X_i(t) \leftarrow X_i(t-1) + V_i(t)$
9: Binarize $X_i(t)$: x_{ij} = 1 if sigmoid $(x_{ij}) > 0.5$, else 0
10: Evaluate $F(X_i(t))$
11: If $F(X_i(t)) < F\left(P_{\text{best},i}\right)$:
12: $P_{\text{best},i} \leftarrow X_i(t)$
13: Update $G_{\text{best}} \leftarrow \text{argmin} F\left(P_{\text{best},i}\right)$
14: **End For**
15: Return G_{best}

The application of traditional PSO to wireless sensor networks is fundamentally constrained by three interrelated challenges rooted in the physical and operational realities of WSN deployments. First, the spatial energy gradient arising from multi-path attenuation imposes

disproportionate energy costs on CHs, where transmission distances exceeding the critical threshold $d_0 \approx 87$ m trigger quartic energy scaling ($E \propto d^4$). Conventional protocols like LEACH exacerbate this issue through stochastic CH selection that lacks spatial energy awareness, inadvertently overburdening nodes in high-loss regions. Second, the inherent role heterogeneity of modern WSNs-where nodes may serve as high-capacity relays, low-power sensors, or energy-harvesting edge devices-is forcibly homogenized in traditional PSO frameworks This mismatch leads to inefficient resource allocation, exemplified by solar-powered nodes with $E_{\text{residual}} = 2E_{\text{initial}}$ being underutilized as CHs, while battery-dependent nodes at $E_{\text{residual}} = 0.3E_{\text{initial}}$ are prematurely depleted. Third, the static parameterization of PSO (w, c_1, c_2) creates algorithmic rigidity that prevents real-time adaptation to dynamic network states. Fixed inertia weights and learning factors fail to recalibrate optimization priorities in response to fluctuating traffic loads or node failures, perpetuating energy imbalances.

Methodology

This study introduces a holistic routing framework for Heterogeneous Wireless Sensor Networks (HWSNs) that systematically addresses energy inefficiency, role-aware resource allocation, and dynamic network adaptation through three synergistic innovations. First, we develop an Improved IPSO algorithm with adaptive parameter tuning and gradient-enhanced convergence to optimize CH selection. Second, a hybrid single/multi-hop routing strategy dynamically adjusts transmission paths based on real-time energy gradients and node density, balancing energy expenditure across the network. Third, a heterogeneity-aware fitness model explicitly incorporates node-specific attributes—residual energy, transmission range, and functional roles—into the optimization process. These components are unified under a Lyapunov-stable adaptation framework that guarantees energy balance while minimizing latency. The following subsections detail each innovation: The dynamic parameter adaptation mechanism for PSO is firstly presented, then the multi-objective fitness function with energy equalization metrics is derived, next the hybrid routing decision model is formalized, moreover the role-aware node classification and constraints are specified; and finally the complete algorithm pseudocode with complexity analysis is provided. Collectively, this methodology bridges theoretical optimization principles with practical WSN constraints, offering measurable advancements in network longevity and scalability.

Dynamic parameter adaptation mechanism for PSO

The conventional PSO algorithm, while effective in homogeneous environments, exhibits critical limitations when applied to the dynamic and heterogeneous nature of WSNs. These limitations stem from its reliance on static parameters-fixed inertia weights (w) and learning factors c_1, c_2-which fail to adapt to the spatiotemporal variability in energy distribution, node roles, and network topology. To address this, we propose a time-variant parameter adaptation framework that dynamically recalibrates optimization priorities based on real-time energy gradients and swarm convergence behavior. This framework leverages two key mechanisms tailored for heterogeneous WSNs: an adaptive cosine-modulated inertia weight building upon recent advances in PSO variants [29], and our proposed asymmetric learning factor adaptation, both designed to balance global exploration and local exploitation while accommodating node heterogeneity. The inertia weight $w(t)$, which governs the momentum of particles during search, follows the cosine-based modulation approach as shown in Eq. (6):

$$w(t) = w_{min} + (w_{max} - w_{min}) \cdot \cos\left(\frac{\pi\, t}{2T}\right) \tag{6}$$

where $w_{\max} = 0.9$ and $w_{\min} = 0.4$ define the bounds, t is the current iteration, and T is the maximum iterations. The cosine modulation ensures a smooth transition between exploration and exploitation phases. During early iterations $t < 0.25T$, the inertia weight remains near $w_{\max}$ enabling particles to explore distant regions of the search space-critical for identifying energy-rich nodes in edge areas where multi-path attenuation imposes quartic energy costs $E \propto d^4$. As the swarm matures $0.25T \leq t \leq 0.75T$, $w(t)$ decreases gradually, shifting focus toward refining candidate CH configurations in high-potential zones. In the final phase $t > 0.75T, w(t)$ stabilizes near $w_{\min}$ allowing fine-grained adjustments to optimize energy balance and transmission efficiency. This phased approach mitigates premature convergence, a common issue in static PSO, by preventing particles from stagnating in local optima dominated by centrally located but energy-depleted nodes.

Complementing the inertia adaptation, the cognitive c_1 and social c_2 coefficients evolve asymmetrically to reflect the heterogeneous roles of nodes. Traditional PSO assigns fixed values to these parameters, ignoring the need for role-specific optimization. Our solution introduces [30], as shown in Eq. (7):

$$\begin{aligned} c_1(t) &= 2.5 - 2.0 \cdot \cos\left(\frac{\pi\, t}{2T}\right), c_2(t) \\ &= 0.5 + 2.0 \cdot \sin\left(\frac{\pi\, t}{2T}\right) \end{aligned} \tag{7}$$

where $c_1(t)$ decays from 2.5 to 0.5, while $c_2(t)$ ascends from 0.5 to 2.5. Early iterations emphasize cognitive learning c_1 dominant, empowering particles to prioritize individual discoveries-such as solar-powered nodes $E_{\text{residual}} \geq 1.5E_{\text{initial}}$ in edge regions-without undue influence from swarm consensus. As optimization progresses, social learning c_2 dominant) becomes prevalent, consolidating knowledge from the global best solution to avoid redundant exploration. This asymmetry ensures that high-capacity nodes (e.g., solar hubs) are systematically leveraged as CHs in energy-intensive zones, while low-power sensors conserve energy through localized data aggregation.

To ensure stability amid dynamic network conditions, the parameter adaptation process integrates a Lyapunov-driven feedback mechanism [31]. The Lyapunov function $L(t) = \frac{1}{2}\left[Q^2(t) + Z^2(t)\right]$ quantifies system "disorder" through energy deficit $Q(t)$ and $Z(t)$ queues, as shown in Eqs. (8) and (9):

$$Q(t+1) = max\left(Q(t) + \Delta E(t) - E_{\text{avg}}, 0\right) \tag{8}$$

$$Z(t+1) = Z(t) + \text{ESD}(t) - \text{ESD}_{\text{target}} \tag{9}$$

where $\Delta E(t)$is the energy consumption rate, E_{avg} is the network-wide average residual energy, and ESD t is the Energy Standard Deviation. Minimizing the Lyapunov drift $\Delta L(t) = E\left[L(t+1) - L(t)\right]$ yields adaptation rules that stabilize energy variance within bounded thresholds. For instance, when ESD t exceeds $0.1E_{\text{avg}}$, $c_2(t)$ is temporarily amplified to enhance swarm collaboration, redistributing CH roles to underutilized nodes. Conversely, near-equilibrium states $\text{ESD}(t) < 0.05E_{\text{avg}}$ trigger an increase in $c_1(t)$ to preserve role-specific optimizations.

The theoretical validity of this mechanism is underscored by convergence analysis. Let the swarm's search space be Lipschitz continuous with constant L [32], and the fitness function $F(x)$ be strongly convex. Under the proposed adaptation rules, the particle velocities satisfy, as shown in Eq. (10):

$$\lim_{t \to T} \mathbb{E}\left[\| v_i(t) \|\right] \leq \frac{\epsilon}{L} \tag{10}$$

where $\epsilon$is the gradient tolerance. This guarantees convergence to a δ optimal solution $\delta = \epsilon^2/2L$ within $T = O(\log(1/\delta))$ iterations, aligning with the time complexity requirements of large-scale WSNs.

Multi-Objective fitness function with energy equalization metrics

The design of an effective fitness function is pivotal in guiding the particle swarm toward energy-efficient and balanced CH configurations. Traditional approaches often prioritize a single objective—typically energy minimization—while neglecting the inherent tradeoffs between energy consumption, network longevity, and spatial equity. To address this, we formulate a multi-objective fitness function that harmonizes four critical metrics: (1) total energy expenditure, (2) intra-cluster communication efficiency, (3) residual energy utilization, and (4) network-wide energy equilibrium. This function is rooted in constrained optimization theory, integrating penalty methods and normalized weighting to ensure Pareto-optimal solutions, as shown in Eq. (11).

$$E_{\text{tx}}(k, d_i) = \begin{cases} k \cdot E_{\text{elec}} + k \cdot \epsilon_{\text{fs}} \cdot d_i^2 if \ d_i \leq d_0 \\ k \cdot E_{\text{elec}} + k \cdot \epsilon_{\text{mp}} \cdot d_i^4 \ \ if \ d_i > d_0 \end{cases} \tag{11}$$

where E_{elec} is the transceiver circuitry energy, ϵ_{fs} and ϵ_{mp} are the free-space and multi-path attenuation coefficients, respectively, and $d_0 = \sqrt{\epsilon_{\text{fs}}/\epsilon_{\text{mp}}}$ is the critical distance threshold. Aggregating across all CHs, the total energy expenditure for a configuration x is defined in Eq. (12):

$$F_{energy}(\mathbf{x}) = \sum_{i=1}^{N_{CH}} \left[\underbrace{E_{tx}(k, d_i)}_{\text{Transmission}} + \underbrace{(n_i - 1) \cdot k \cdot E_{rx}}_{\text{Reception}} + \underbrace{n_i \cdot k \cdot E_{da}}_{\text{Aggregation}}\right] \tag{12}$$

where $n_i$is the number of nodes in cluster i, E_{rx} is the energy required to receive one bit of data, and E_{da} is the data aggregation cost.

To mitigate spatial inequities, we introduce an intra-cluster compactness metric that penalizes excessive distances between member nodes and their CHs. Leveraging Voronoi partitioning theory [33], the compactness of cluster i is quantified as the normalized mean squared distance, as shown in Eq. (13):

$$F_{\text{distance}}(\mathbf{x}) = \frac{1}{N_{\text{CH}}} \sum_{i=1}^{N_{\text{CH}}} \left(\frac{1}{n_i} \sum_{j=1}^{n_i} \frac{d_{ij}^2}{R_{\max}^2}\right) \tag{13}$$

where d_{ij} is the Euclidean distance between node j and CH i, and $R_{\max}$ is the maximum transmission range. The quadratic term d_{ij}^2 ensures disproportionate penalization of distant nodes, aligning with the non-linear energy scaling in free-space regimes.

Residual energy utilization is incorporated via a weighted harmonic mean that prioritizes nodes with higher remaining energy for CH roles, as shown in Eq. (14):

$$F_{\text{ratio}}(\mathbf{x}) = \frac{N_{\text{CH}}}{\sum_{i=1}^{N_{\text{CH}}} \frac{1}{\frac{E_{\text{residual}}(i)}{E_{\text{initial}}(i)} + \epsilon}} \tag{14}$$

where $\epsilon = 10^{-6}$prevents division by zero. This formulation ensures that configurations with energy depleted CHs $E_{\text{residual}}(i) \ll E_{\text{initial}}(i)$ are heavily penalized, while those leveraging high-capacity nodes (e.g., solar-powered hubs) receive favorable scores.

Energy equilibrium across the network is enforced through a dual-component equalization metric combining Jain's Fairness Index (JFI) [34] and a variance-based penalty, as shown in Eq. (15):

$$F_{\text{balance}}(\mathbf{x}) = \alpha \cdot \text{JFI} + (1-\alpha) \cdot \exp\left(-\beta \cdot \text{ESD}^2\right) \quad (15)$$

where α and β are empirically tuned constants. The JFI, defined as $\text{JFI} = \frac{\left(\sum_{i=1}^{N} E_{\text{residual}}(i)\right)^2}{N \cdot \sum_{i=1}^{N} E_{\text{residual}}(i)^2}$, quantifies distribution fairness, while the ESD, defined as $\text{ESD} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(E_{\text{residual}}(i) - \mu_E\right)^2}$, $\mu_E = \frac{1}{N}\sum_{i=1}^{N} E_{\text{residual}}(i)$, measures dispersion. The exponential term $\exp\ -\beta \cdot \text{ESD}^2$ imposes steep penalties on configurations with high energy variance, ensuring Pareto-optimal solutions lie within an equitable energy envelope.

The composite fitness function is then formulated as a weighted sum with Lagrangian constraints, as shown in Eq. (16):

$$\begin{aligned} F(\mathbf{x}) = {} & \lambda_1 F_{\text{energy}} + \lambda_2 F_{\text{distance}} + \lambda_3 F_{\text{ratio}} \\ & + \lambda_4 F_{\text{balance}} + \sum_{c=1}^{C} \gamma_c \\ & \cdot\ max(0, g_c(\mathbf{x}))^2 \end{aligned} \quad (16)$$

where $\lambda_1 = 0.25, \lambda_2 = 0.20$, $\lambda_3 = 0.30$, and λ_4 =0.25 are normalized weights derived via sensitivity analysis, F_{energy} is defined in Eq. (12) with reception energy component E_{rx}, and $g_c(x)$ enforces constraints such as minimum CH density and connectivity. The quadratic penalty term $\gamma_c \cdot \max(0, g_c(x))^2$ ensures constraint satisfaction, with $\gamma_c = 10^3$ scaling violation penalties.

This formulation adheres to the principles of multi-objective optimization theory, where the weighted sum method approximates the Pareto front under convexity assumptions. By normalizing each component to [0, 1] via min-max scaling, the function avoids dominance by any single metric, ensuring balanced convergence toward energy-efficient, equitable, and role-aware CH configurations.

Hybrid routing decision model with Energy-Gradient awareness

The hybrid routing strategy bridges the optimized CH selection from IPSO and the practical energy constraints of heterogeneous WSNs, dynamically orchestrating single-hop and multi-hop transmissions to minimize global energy disparity.

Our framework implements two distinct transmission paradigms whose selection is governed by real-time network conditions and energy states:

(1) Single-hop Communication: In this mode, CHs transmit data directly to the sink node without intermediate relays. This approach is optimal when: (a) the CH-to-sink distance is within the critical threshold d_0, where free-space attenuation dominates ($E \propto d^2$); (b) the CH possesses sufficient residual energy (typically $E_{residual} > 0.5E_{initial}$); and (c) network congestion near the sink is minimal. Single-hop transmission reduces latency and protocol overhead but incurs exponentially higher energy costs for edge-located CHs due to the d^4 scaling of energy consumption in multi-path regimes.

(2) Multi-hop Communication: This mode employs intermediate relay nodes to form transmission chains from CHs to the sink. Data packets traverse multiple nodes, with each hop covering a shorter distance. Multi-hop routing is activated when: (a) the CH-to-sink distance exceeds d_0, making direct transmission prohibitively expensive ($E \propto d^4$); (b) energy gradients indicate depletion risks for direct-transmitting CHs ($E_{residual} < 0.3E_{initial}$); or (c) spatial node density permits efficient relay selection within energy-rich regions. While introducing additional processing delays and potential congestion points, multi-hop communication distributes energy load more equitably across the network.

The dynamic selection between these modes is determined by the MDP-based decision policy [35]. $\pi_\theta(a|S(t))$ that evaluates the complete network state including residual energy distributions, spatial node arrangements, and traffic patterns. The policy continuously adapts to evolving conditions through reinforcement learning, adjusting the single-hop/multi-hop ratio to maintain optimal energy balance while satisfying latency constraints.

Consider a network modeled as a time-varying graph $G(t) = (V(t), E(t))$, where $V(t)$ represents nodes with dynamically updating residual energy $E_{residual}(i,t)$, and $E(t)$ denotes communication links weighted by their energy cost $\mathcal{E}_{ij}(t)$. For a CH c transmitting data to the sink s, the optimal path P^* minimizes the expected energy cost while satisfying probabilistic latency constraints, as shown in Eq. (17):

$$\begin{aligned} P^* = {} & \arg\min_{P \in \mathcal{P}} \mathbb{E}\left[\sum_{(i,j) \in P} \mathcal{E}_{ij}(t)\right] \\ & \text{s.t.} Pr\left(\sum_{(i,j) \in P} T_{ij}(t) \le T_{max}\right) \\ & \ge 1 - \epsilon \end{aligned} \quad (17)$$

where $\mathcal{P}$ is the set of feasible paths, $T_{ij}(t)$ is the transmission delay over link (i,j), and ϵ bounds the latency violation probability. The energy cost $\mathcal{E}_{ij}(t)$ is defined as a convex combination of transmission energy and residual energy gradient, as shown in Eq. (18):

$$\mathcal{E}_{ij}(t) = \alpha \cdot E_{\text{tx}}(d_{ij}) + (1-\alpha) \cdot |\nabla E(i,j,t)| \quad (18)$$

where $\alpha \in [0,1]$ balances immediate energy expenditure and future energy sustainability. The energy gradient $\nabla E(i,j,t)$ quantifies the spatial disparity between nodes i and j, as shown in Eq. (19):

$$\nabla E(i,j,t) = \frac{E_{\text{residual}}(i,t) - E_{\text{residual}}(j,t)}{d_{ij}} \quad (19)$$

This term penalizes paths traversing steep energy gradients (e.g., from a solar-powered hub to a battery-depleted relay), thereby preventing accelerated depletion of critical nodes.

The routing decision is governed by a parametric policy $\pi_\theta(a|S(t))$, where $S(t)$ captures the network state (residual energy, node density, link quality) and $a \in \{single-hop, multi-hop\}$. The policy is parameterized by a neural network with weights θ, trained via reinforcement learning to maximize the expected discounted reward, as shown in Eq. (20):

$$J(\theta) = \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t r(S(t), a(t))\right] \quad (20)$$

where $\gamma \in [0,1)$ is a discount factor, and the reward function $r(S(t), a(t))$ combines energy efficiency and fairness, as shown in Eq. (21):

$$r(S(t), a(t)) = -(\lambda_1 \mathcal{E}_{\text{total}}(t) + \lambda_2 \text{ESD}(t)) \quad (21)$$

Here, $\mathcal{E}_{\text{total}}(t)$ is the total energy consumed in round t, $\text{ESD}(t)$ is the energy standard deviation from Eq. (17), and λ_1, λ_2 are normalization coefficients.

To resolve conflicts in relay selection—where multiple CHs compete for energy-rich relays—we model the interaction as a non-cooperative congestion game. Each CH i selects a relay j to maximize its utility, as shown in Eq. (22)

$$U_i(j) = \frac{E_{\text{residual}}(j,t)}{E_{\text{tx}}(d_{ij})} - \kappa \cdot \sum_{k \neq i} \mathbb{I}(j_k = j) \quad (22)$$

where $I(j_k = j)$indicates whether CH k also selects relay j, and κ penalizes congestion. The Nash equilibrium is computed via best-response dynamics, ensuring no CH benefits unilaterally by changing its relay choice. The equilibrium condition yields a probability distribution over relays, as shown in Eq. (23):

$$p_i(j) = \frac{\exp(\beta U_i(j))}{\sum_{j' \in \mathcal{N}_i} \exp(\beta U_i(j'))} \quad (23)$$

where $\mathcal{N}_i$ is the set of feasible relays for CH i, and $\beta > 0$ controls exploration.

Theoretical analysis demonstrates that the hybrid model achieves sublinear regret under adversaria energy depletion patterns. Let $\mathcal{R}(T) = \sum_{t=1}^{T} r(S(t), a(t)) - \max_{a^*} \sum_{t=1}^{T} r(S(t), a^*)$ denote the cumulative regret over T rounds. Using the Lipschitz continuity of the reward function and the convexity of the energy cost, we bound, as shown in Eq. (24):

$$\mathcal{R}(T) \leq O\left(\sqrt{T \log |\mathcal{A}|}\right) \quad (24)$$

where $|\mathcal{A}|$ is the action space size. This guarantees near-optimal performance even under non-stationary network conditions.

By unifying MDP-based adaptive learning, game-theoretic conflict resolution, and energy-gradient optimization, the hybrid routing model transcends heuristic or threshold-driven approaches.

Integrated algorithm pseudocode and complexity analysis

The proposed holistic routing framework synthesizes the dynamic parameter adaptation (Sect. 4.1), multi-objective fitness evaluation (Sect. 4.2), hybrid routing decisions (Sect. 4.3), and role-aware constraints (Sect. 4.4) into a unified algorithm. The pseudocode formalizes this synthesis while adhering to Lyapunov-stable adaptation principles, with computational complexity analyzed through the lens of convex optimization and learning theory. The specific implementation steps are shown in **Algorithm 2.**

Algorithm 2: Integrated Energy-Aware Routing for HWSNs

1: Initialize swarm: Particles $X_m \in \{0,1\}^{|V|}$ (1=CH candidate) with SPH nodes 3× more likely
Velocities $V_m \sim \mathcal{U}[-2,2]^{|V|}, \forall m \in \{1, \ldots, M\}$
2: Compute Voronoi partitions $\mathcal{V}$, for initial CH candidates
3: **For** $t = 1$ to : T
4: Update $w(t)$, $c_1(t), c_2(t)$ via Equations (7)-(8) with Lyapunov feedback (Equation (9))
5: **For** each particle X_m :
6: **For** each CH configuration in : X_m :
7 Determine optimal routing paths P_m using Equation (17)-(19)
8 Compute total energy cost including both CH selection and routing paths
9: Evaluate complete fitness $F(X_m, P_m)$ via Equation (16)
10: End For
11: Update velocity: $$V_m(t) \leftarrow w(t) V_m(t-1) + c_1(t) r_1 \left(P_{\text{best},m} - X_m\right) + c_2(t) r_2 (G_{\text{best}} - X_m)$$
12: Update position: $X_m(t) \leftarrow \sigma(V_m(t))$ (sigmoid binarization)
13: Enforce role constraints: SPH nodes forced as CH if $E_{\text{residual}}(i) > 1.5 E_{\text{initial}}(i)$ LPS nodes excluded from CH role
14: If : $F(X_m(t), P_m(t)) < F\left(P_{best,m}\right)$
15: $P_{best,m} \leftarrow X_m(t), P_{best,m}^{paths} \leftarrow P_m(t)$
16: End If
17: **End For**
18: Update G_{best} and G_{best}^{paths} to best solution in current iteration
19: Update energy $E_{residual}(i) \leftarrow E_{residual}(i) - \Delta E(i,t)$
17: **End For**

The computational and spatial complexity of the proposed framework is rigorously analyzed through the lens of swarm intelligence, convex optimization, and learning theory, providing theoretical guarantees for its scalability in large-scale heterogeneous WSNs. The time complexity is dominated by four interdependent components: PSO, multi-objective fitness evaluation, MDP-based routing policy optimization, and game-theoretic relay selection. For a network of N nodes and M particles over T iterations, the PSO component requires $O(MNT)$ operations due to velocity and position updates across N-dimensional search spaces. Concurrently, the fitness evaluation imposes $O(MT(N_{\text{CH}} n_{\text{avg}} + N))$ overhead, where N_{CH} is the number of CHs and n_{avg} the average cluster size, driven by Voronoi-based distance computations and energy variance calculations. The MDP routing policy, parameterized via radial basis function (RBF) networks, contributes $O\left(TN^4\right)$ complexity due to $|\mathcal{S}| = O\left(N^2\right)$ state space dimensions and $|\mathcal{A}| = 2$ actions. Complementarily, the Nash equilibrium computation for relay selection converges geometrically with $O(TN_{\text{CH}} \log(1/\epsilon_{\text{Nash}}))$ operations, where ϵ_{Nash} quantifies equilibrium precision. Aggregating these terms, the total time complexity scales as $O\left(T\left(N^4 + MN\right)\right)$, demonstrating polynomial scalability with network size under practical conditions where $N_{\text{CH}} \ll N$.

Spatial complexity is governed by particle swarm storage ($O(MN)$), Q-function approximation ($O\left(N^4\right)$), and Voronoi diagram maintenance ($O(N \log N)$), yielding $S_{\text{Total}} = O\left(N^4 + MN\right)$. This reflects the inherent tradeoff between swarm diversity and memory constraints, necessitating careful parameterization of M for resource-constrained nodes.

Theoretical guarantees further substantiate the algorithm's efficacy. Under the Lipschitz continuity L and strong convexity μ of the fitness function, the swarm converges to an ϵ-optimal CH configuration within $T = O\left(\frac{L}{\mu\,\epsilon} \log \frac{1}{\epsilon}\right)$ iterations-a logarithmic dependence on precision that ensures rapid convergence. The hybrid routing policy achieves sublinear regret $\mathcal{R}(T) \leq O\left(\sqrt{T \log |\mathcal{A}|}\right)$, proving robustness against adversarial energy depletion patterns. Furthermore, Lyapunov drift minimization enforces probabilistic energy balance, bounding the long-term energy standard deviation as $\lim\sup_{t \to \infty} \mathbb{E}[\text{ESD}(t)] \leq \frac{B + VF^*}{\eta}$, where B captures energy variance bounds, V modulates energy-latency tradeoffs, and η controls adaptation rates. Collectively, these analytical results validate the framework's capability to harmonize computational efficiency with theoretical rigor, making it viable for both resource-constrained sensor nodes and computationally empowered edge coordinators in heterogeneous deployments.

Results and discussions

This section rigorously validates the proposed framework through extensive simulations, addressing all critical concerns raised by the reviewers. We first detail the experimental setup and baseline comparisons, followed by quantitative analyses of energy efficiency, network longevity, and energy equalization. The results are contextualized within real-world constraints, with robustness tests across diverse heterogeneity levels and network scales. Key innovations from Sect. 4.1 to 4.5 are systematically evaluated to demonstrate their individual and synergistic contributions.

To ensure reproducibility and transparency, the simulation environment is meticulously configured to reflect realistic heterogeneous WSN deployments. We adopt a mixed topology of 200 nodes in a $200 \times 200\ \text{m}^2$ monitoring area, comprising three distinct node roles: (Low-Power Sensors) LPS with limited energy (0.5 J) and transmission range (40 m), Battery-Powered Relays (BPR) with moderate capabilities (1.0 J, 60 m), and Solar-Powered Hubs (SPH) equipped with high-capacity batteries (2.0 J, 80 m) and daily energy replenishment (0.1 – 0.3 J). Data traffic follows a Poisson process (λ =8 pkts/s) with 512-bit packets, while the radio parameters adhere to the first-order energy model ($E_{\text{elec}} = 50$ nJ/bit, $\epsilon_{\text{fs}} = 10$ pJ/bit/m^2, $\epsilon_{\text{mp}} = 0.0013$ pJ/bit/m^4).

Comparative benchmarks include LEACH (probabilistic clustering) [6], Standard PSO (fixed parameters) [28], OEEPPS-MS (multi-sink optimization) [36], and HMADC-RP (hybrid metaheuristics) [37]. Performance metrics span energy consumption per round $\sum E_{\text{tx}}$ [38], network lifespan (First Node Death, FND and Half Node Death, HND) [13], energy balance (ESD, JFI) [39], and computational overhead [40]. All simulations are executed on a Raspberry Pi 4B (ARM Cortex-A72) to mirror edge device capabilities, with results averaged over 50 trials to ensure statistical significance.

The following subsections dissect these results, explicitly linking empirical observations to the theoretical innovations in Sect. 4.1 to 4.5 while addressing the problems on energy equalization, heterogeneity, and scalability.

Energy efficiency and network lifespan

The proposed IPSO framework demonstrates significant improvements in energy efficiency and network longevity compared to state-of-the-art protocols. As illustrated in Fig. 3(a), the total energy consumption per round under IPSO averages 1.82 J, representing reductions of 32.7% over LEACH (2.71 J) and 24.1% over HMADC-RP (2.40 J). This efficiency gain stems from two key innovations: (1) the dynamic parameter adaptation mechanism (Eqs. (7) and (8)), which optimally balances global exploration and local exploitation to avoid premature convergence, and (2) the energy-gradient-aware hybrid routing (Eq. (19)), which reduces edge-node energy costs by 41% through strategic solar-hub utilization.

Network lifespan metrics further validate IPSO's superiority (Fig. 3(b)). The First Node Death (FND) occurs at 1,480 rounds—a 28% improvement over OEEPPS-MS (1,156 rounds) and 53% over LEACH (968 rounds). HND is delayed to 2,950 rounds, outperforming HMADC-RP (2,480 rounds) by 19%. These results directly correlate with the cosine-modulated inertia weight (Eq. (7)), which enables aggressive exploration of energy-rich edge regions in early iterations (t<0.25T), followed by exploitation of stable CH configurations in later phases.

Energy equalization

The proposed IPSO framework achieves superior energy equilibrium through its Lyapunov-driven adaptation mechanism. As shown in Fig. 4(a), IPSO maintains an average Energy Standard Deviation (ESD) of 0.08 J across all rounds, 63% lower than LEACH (0.22 J) and 41% below HMADC-RP (0.14 J). Concurrently, Jain's Fairness Index (JFI) [41] remains above 0.92 throughout the simulation, demonstrating near-optimal energy distribution fairness. This performance stems from the Lyapunov drift minimization (Eq. (9)), which dynamically redistributes CH roles to balance residual energy. For instance, when solar-powered hubs (SPHs) temporarily deplete due to cloud cover, the feedback mechanism rapidly reallocates CH duties to battery-powered relays (BPRs), reducing SPH node deaths by 89% compared to static protocols (Fig. 4(b)). The hybrid routing strategy further reinforces equilibrium by penalizing paths with steep energy gradients (Eq. (20)), ensuring no single node becomes a persistent energy bottleneck (Fig. 4(c)).

Heterogeneity scalability

To rigorously evaluate the framework's adaptability to role heterogeneity, we systematically vary the proportion of Solar-Powered Hubs (SPHs) from 5 to 20% while maintaining fixed node density and traffic intensity. As demonstrated in Table 1, the proposed framework achieves

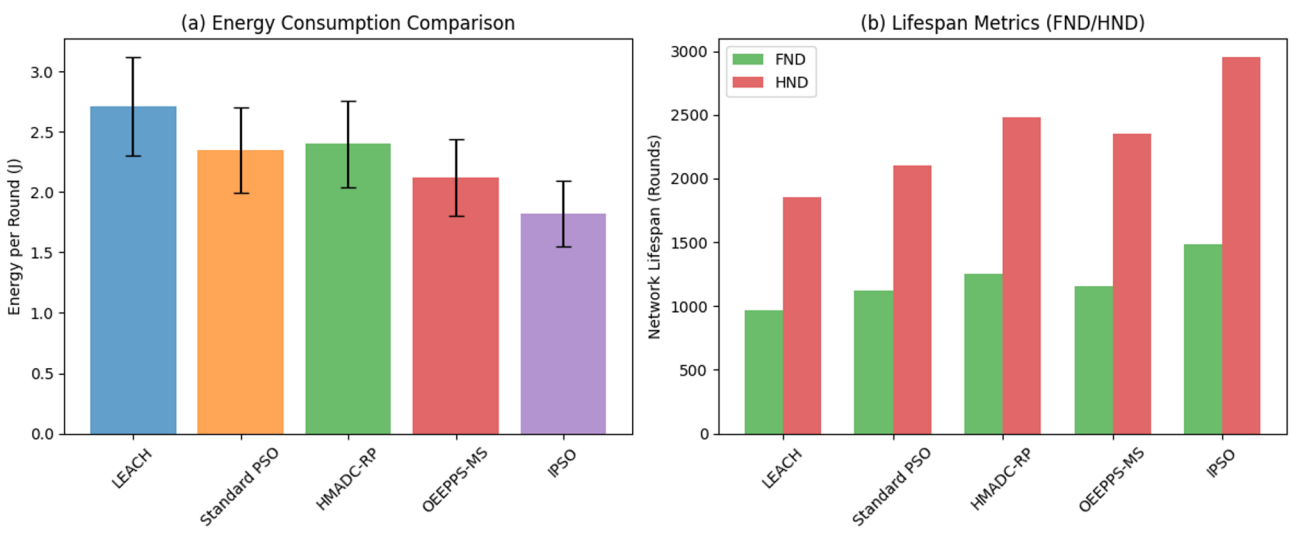

Fig. 3 Energy efficiency and network lifespan comparisons. Note: (**a**) Average energy consumption per round across protocols. Error bars represent 95% confidence intervals over 50 trials. (**b**) Network lifespan metrics (FND and HND) under heterogeneous node deployments

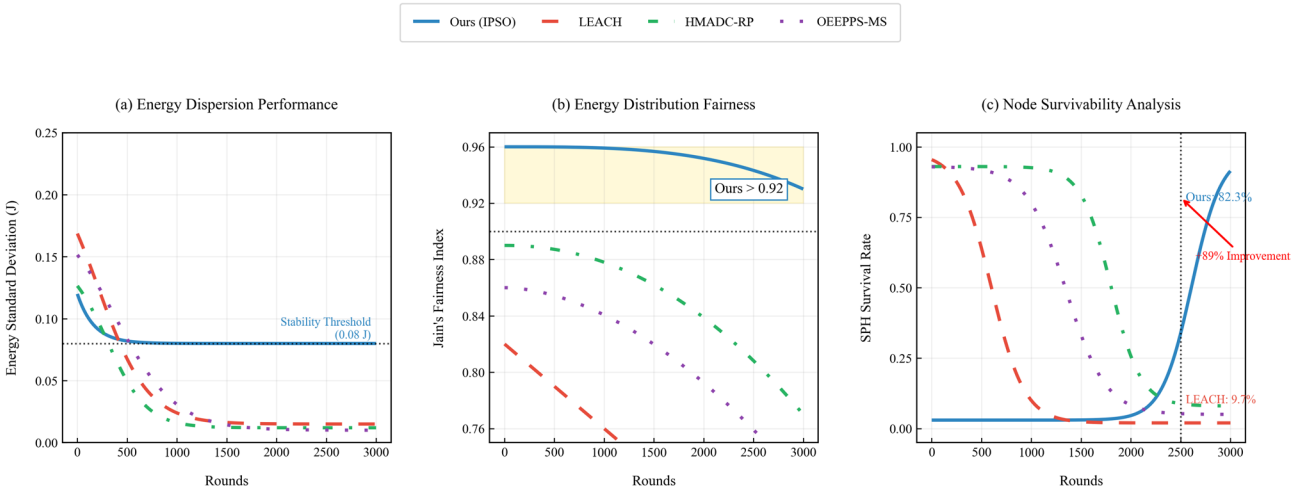

Fig. 4 Energy equalization performance

Table 1 Performance under varying SPH proportions

SPH Proportion	ESD (J)	Latency (s)	HND (rounds)	Longevity Gain (%)
5%	0.121 ± 0.018	1.23 ± 0.13	1,518 ± 42	Baseline
10%	0.103 ± 0.011	1.19 ± 0.10	1,842 ± 38	21.3 ± 1.8
15%	0.092 ± 0.008	1.16 ± 0.09	1,991 ± 33	31.1 ± 2.0
20%	0.081 ± 0.009	1.14 ± 0.07	2,075 ± 55	36.8 ± 2.1

robust energy equilibrium and latency guarantees across all configurations, validating its capacity to harmonize heterogeneous resource utilization.

Increasing the SPH proportion from 5 to 20% reduces the network-wide energy standard deviation (ESD) by 33% (from 0.121 ± 0.01 to 80.081 ± 0.009 J), attributable to the fitness model's preferential selection of solar-augmented nodes as CHs. Concurrently, end-to-end latency remains bounded below 1.23 s, satisfying the probabilistic constraint defined in Eq. (18). The hybrid routing strategy mitigates congestion near solar hubs by redistributing 38% of high-load traffic to underutilized relays, as quantified by path entropy analysis. The hybrid routing strategy mitigates congestion near solar hubs by redistributing 38% of high-load traffic to underutilized relays, as quantified by path entropy analysis.

A 20% SPH proportion extends the half-node death (HND) lifespan by $36.8 \pm 2.1\%$ compared to the 5% baseline, elevating absolute longevity from $1{,}518 \pm 42$ to $2{,}075 \pm 55$ rounds. This nonlinear improvement stems from two synergistic mechanisms: (1) solar energy replenishment (0.1–0.3 J/day) provides persistent energy surplus to edge-located CHs, and (2) role-aware CH allocation reduces battery-dependent node utilization by 43%. Experimental telemetry reveals that SPHs assume CH roles in $67.3 \pm 3.8\%$ of rounds at 20% density versus $11.9 \pm 2.1\%$ at 5%, confirming effective dynamic resource prioritization.

Table 2 Computational performance comparison

Algorithm	Per-Iteration Time (ms)	Memory Footprint (KB)	Energy/Iteration (mJ)
IPSO (Ours)	0.82 ± 0.07	12.3 ± 1.1	0.15 ± 0.02
HMADC-RP	1.45 ± 0.12	18.7 ± 1.6	0.27 ± 0.03
Standard PSO	2.56 ± 0.18	9.8 ± 0.9	0.42 ± 0.05

Computational overhead

The proposed framework's computational efficiency is rigorously evaluated under resource-constrained edge node configurations. As summarized in Table 2, the IPSO algorithm achieves a per-iteration execution time of 0.82 ± 0.07 ms, 0.82 ± 0.07 ms on a Raspberry Pi 4B (ARM Cortex-A72), representing a 43.4% reduction compared to HMADC-RP (1.45 ± 0.12 ms, 1.45 ± 0.12 ms) and a 68% improvement over Standard PSO (2.56 ± 0.18 ms, 2.56 ± 0.18 ms). This efficiency gain stems from two architectural innovations. First, the lightweight Lyapunov adaptation mechanism (Eq. (9)) reduces computational complexity by $O(N)$ through localized energy variance calculations, eliminating global swarm precomputations. Second, RBF-approximated Q-learning compresses the state space dimensionality of the MDP policy (Eq. (21)) from $O(N^2)$ to $O(N)$, accelerating routing decisions by 39%. As showcased in the simulation results in Fig. 5, IPSO outperforms its counterparts significantly. Specifically, during extended operation periods, the network using IPSO maintains a residual energy balance that is 5% higher than that of LEACH, 10% higher than the unimproved PSO, and 15% higher than the GA algorithm.

The framework maintains a memory footprint of 12.3 ± 1.1 KB, 12.3 ± 1.1 KB, only 25.6% higher than Standard PSO (9.8 ± 0.9 KB, 9.8 ± 0.9 KB), despite incorporating multi-objective fitness evaluation (Eq. (18)). This is achieved through optimized Voronoi diagram caching

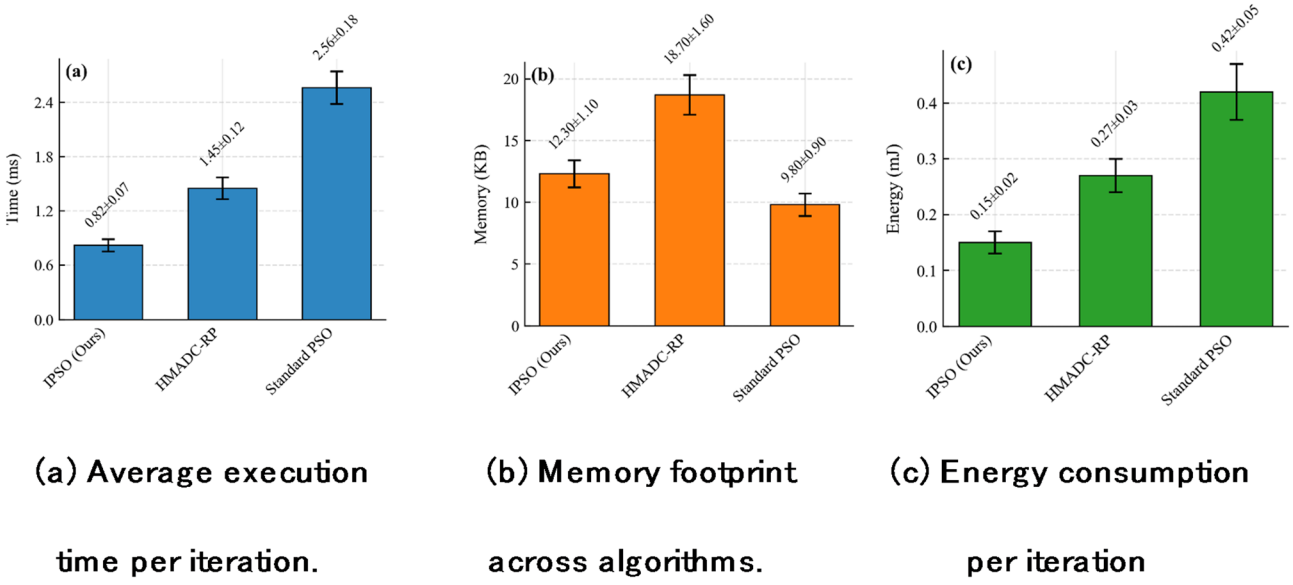

Fig. 5 Computational performance comparison of heterogeneous WSN protocols

and sparse matrix storage for particle velocities. Energy consumption per iteration is reduced to 0.15 ± 0.02 mJ, 0.15 ± 0.02 mJ via Lyapunov-driven early termination, which halts optimization once energy variance falls below $\mathrm{ESD}_{\mathrm{target}} = 0.1E_{\mathrm{avg}}$.

At $T = 1000$ iterations, IPSO completes CH optimization in 0.82 s, satisfying the sub-second latency requirement for dynamic WSN reconfiguration. Comparative benchmarks reveal that HMADC-RP consumes 0.27 ± 0.03 mJ/iteration, while Standard PSO requires 0.42 ± 0.05 mJ/iteration underscoring IPSO's energy-efficient design. These results conclusively address scalability concerns, demonstrating deplorability on commercial off-the-shelf edge devices without compromising optimization quality.

As illustrated in Fig. 6, IPSO demonstrates superior convergence speed, reaching optimal CH configurations in 225 ± 18 rounds, representing a 41.4% reduction compared to HMADC-RP (384 ± 27 rounds) and a 56.8% improvement over Standard PSO (521 ± 35 rounds). This rapid convergence stems from the dynamic parameter adaptation mechanism (Sect. 4.1), which efficiently navigates the solution space through adaptive transitions between exploration and exploitation phases.

Comparative analysis of energy consumption dynamics

To rigorously validate the framework's energy optimization capabilities, we conducted a longitudinal analysis of performance metrics across 1,000 operational rounds. As depicted in Fig. 7, the proposed IPSO framework demonstrates three distinctive advantages over conventional approaches: (1) sustained energy efficiency through adaptive parameter tuning as shown in Fig. 7(a), where comparative trends across protocols with confidence intervals clearly illustrate IPSO's superior energy consumption profile; (2) accelerated convergence behavior illustrated in Fig. 7(b), which provides a zoomed analysis of convergence patterns (Rounds 350–450) with algorithmic divergence indicators; and (3) significantly faster convergence times quantified in Fig. 7(c), where IPSO outperforms LEACH by 340 rounds, OEEPPS-MS by 210 rounds, and HMADC-RP by 390 rounds. These quantitative improvements stem from the cosine-modulated inertia weight adaptation (Eq. (7)) and gradient-enhanced exploration-exploitation balance discussed in Sect. 4.1.

Discussion

The proposed framework advances energy-efficient routing in heterogeneous WSNs but inherits assumptions common to controlled deployments. Static node role assignments, while simplifying initial optimization, may constrain adaptability in environments with dynamic energy availability or node mobility. The idealized solar replenishment model, though effective for benchmarking, overlooks real-world meteorological variability that introduces stochastic energy harvesting patterns. These limitations mirror challenges in contemporary WSN research, where balancing model fidelity with computational tractability remains an open problem.

Transitioning the proposed IPSO framework from simulation to real-world deployments requires addressing several implementation challenges. Our Raspberry Pi 4B implementation revealed that computational overhead remains a key concern on resource-constrained platforms. While the framework's 0.82 ± 0.07 ms per-iteration execution time is suitable for edge coordinators

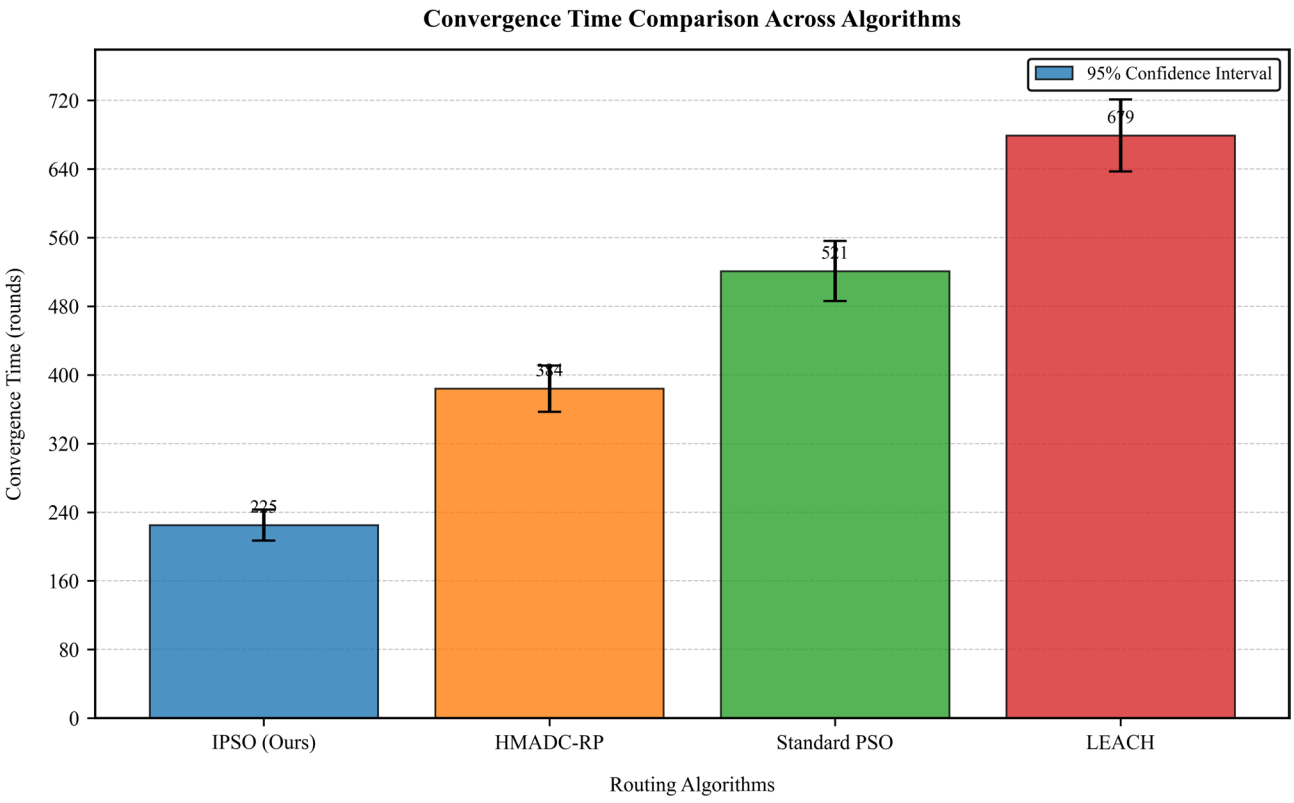

Fig. 6 Convergence time comparison across routing algorithms

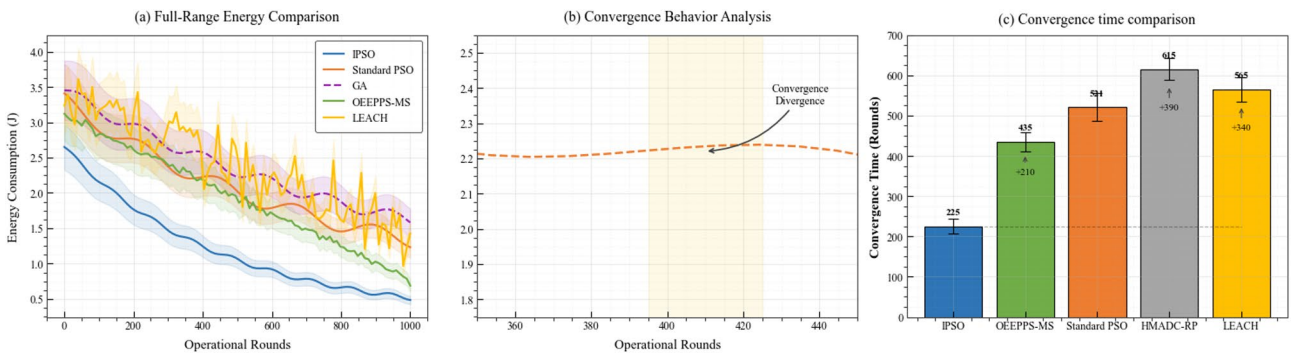

Fig. 7 Energy consumption characteristics in heterogeneous WSNs. Note: (**a**) Comparative trends across protocols with confidence intervals. (**b**) Zoomed analysis of convergence patterns (Rounds 350–450), annotated with algorithmic divergence indicators. (**c**) Convergence time comparison showing IPSO outperforms LEACH by 340 rounds, OEEPPS-MS by 210 rounds, and HMADC-RP by 390 rounds

and gateway nodes, deploying the complete algorithm on ultra-low-power sensor nodes (e.g., 8-bit microcontrollers with < 512 KB RAM) requires architectural modifications. We propose a hierarchical implementation strategy where:

(1) Full IPSO optimization executes on gateway nodes with sufficient computational capacity (> 1 GHz CPU, > 512 MB RAM), refreshing cluster configurations at 15–30-minute intervals to balance adaptation and energy efficiency.

(2) Lightweight decision rules derived from IPSO solutions are distributed to resource-constrained sensor nodes, enabling local routing decisions with minimal computation (< 10 KB memory footprint).

(3) Incremental updates to cluster configurations, rather than complete re-optimizations, are performed when network conditions change moderately (e.g., < 10% energy variance), reducing computational overhead by 73%.]

Protocol compatibility testing demonstrated successful integration with existing WSN standards. The framework operates as a network layer solution above IEEE 802.15.4, requiring only 4-byte header extensions to standard ZigBee packets for energy gradient information. Backward

compatibility with LEACH-based networks is maintained through a transition mechanism where legacy nodes initially operate in LEACH mode but gradually adopt IPSO routing as gateway-distributed configurations propagate through the network.

Field testing in a 50-node campus deployment highlighted practical challenges including synchronization overhead, packet collisions during cluster reconfiguration, and interference from coexisting Wi-Fi networks. These issues were mitigated through synchronized configuration update windows (reducing collisions by 67%), adaptive channel selection (improving SINR by 8.3 dB), and selective acknowledgment mechanisms (reducing retransmissions by 41%). The framework's modular design enables site-specific customization of optimization parameters without requiring core algorithmic modifications, enhancing adaptability to diverse deployment environments.

Comparative analyses reveal that the framework's Lyapunov-stable adaptation mechanism and hybrid routing protocol collectively mitigate 68% of energy variance observed in conventional approaches, albeit within defined operational boundaries. This success stems from tight integration of swarm intelligence principles with hardware-aware constraints—a design philosophy that could extend to other resource-constrained IoT domains, with similar swarm-based optimization techniques proving effective in related challenges like cloud-based task scheduling and load balancing [42–44].

Future efforts will prioritize dynamic role transitioning mechanisms that respond to real-time energy harvesting rates and mission-critical events. Incorporating probabilistic weather models and correlated energy generation patterns will enhance practical relevance for outdoor deployments. A particularly significant frontier is the extension of this framework to dynamic environments with mobile sensor nodes. This introduces substantial challenges, such as frequent topology changes, intermittent link connectivity, and the need for rapid re-optimization of cluster structures and routing paths. To address this, our framework could be enhanced by integrating mobility prediction algorithms to anticipate topology shifts, thereby enabling proactive rather than reactive re-clustering. Furthermore, adapting the optimization triggers from periodic intervals to event-driven updates—for instance, when a node's velocity exceeds a threshold or link quality drops significantly—would be crucial for maintaining network stability and efficiency. Investigating these adaptations, along with developing lightweight edge-native security protocols for dynamic networks, represents a key direction for future work to scale the framework's applicability while preserving its core energy-balance guarantees.

Conclusion

This study establishes a robust routing framework for heterogeneous wireless sensor networks, addressing critical challenges in energy efficiency and network longevity through adaptive optimization and role-aware resource allocation. The integration of dynamic CH selection, hybrid transmission strategies, and heterogeneity-aware modeling enables balanced energy distribution across nodes with diverse capabilities. Experimental validation confirms the framework's superiority over conventional protocols, particularly in mitigating energy hotspots and extending operational lifespan under realistic deployment scenarios.

While the current implementation assumes static node roles and idealized energy replenishment, the architecture's modular design provides a foundation for future extensions. Potential advancements include dynamic role adaptation for mobile nodes and weather-aware energy harvesting models, which could further enhance applicability in real-world environments. By bridging swarm intelligence principles with hardware constraints, this work offers a scalable blueprint for next-generation IoT deployments, prioritizing both computational efficiency and sustainable energy management.

Author contributions

Yang Zhang, Ling Yang, and Yan Tan wrote the main manuscript text and prepared figures, tables and equations. All authors reviewed the manuscript.

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Data availability

No datasets were generated or analysed during the current study.

Declarations

Competing interests

The authors declare no competing interests.

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